Questions and Answers

Locomotive Air Brake Inspection.

(*Continued from page 226, July,* 1918.)

410. Q.—Can this brake be released with the straight air valve after an application with the automatic brake valve?

A.—Not with the S-3 brake valve, but it can be with an S-3-A brake valve.

411. Q.—After having made the 27 lbs. brake pipe reduction during the test, is the brake valve allowed to remain in release position as when inspecting the E. T. brake?

A.—Yes.

412. Q.—Why?

A.—To make a comparison of the air gauge hands.

413. Q.—What is the next brake valve movement?

A.—Same as for the E. T. equipment, the brake pipe is reduced to 110 lbs. and the rate of equalizing reservoir reduction timed from 110 to 90 lbs.

414. Q.—Is this time the same for the G-6 and H-6 brake valves?

A.—Yes.

415. Q.—What is the time if the equalizing reservoir pressure is 70 lbs. and a 20 lb. reduction is made as with the engine in freight service?

A.—9 to 11 seconds.

416. Q.—Are these two applications sufficient for the automatic brake valve test?

A.—Yes.

417. Q.—What test is made with the straight air brake valve?

A.—An application to see that the brake will be instantly applied, and a movement to release position to see that the release is prompt.

418. Q.—What would cause a blow at the exhaust port of a driver brake triple valve when both brake valves are in release position?

A.—A leaky triple valve slide valve.

419. Q.—What would cause the blow only at a time the straight air brake is applied?

A.—A leak past the automatic side of the driver brake double check valve.

420. Q.—What would cause a blow at the exhaust port of the straight air brake valve when both brakes are released?

A.—A leaky application valve in the straight air brake valve.

421. Q.—If the straight air brake was applied?

A.—A leaky exhaust valve or a leaky handle shaft washer.

422. Q.—What would cause a leak here only at a time the automatic brake was applied?

A.—A leak from the straight air side of one of the double check valves.

423. Q.—Can you tell which one would be at fault?

A.—Yes.

424. Q.—How?

A.—By trying the brake with the tender brake triple valve out, which would eliminate the tender check valve from the disorder.

425. Q.—What would cause a leak at the exhaust port of a quick action triple valve on the tender only at a time the straight air brake was applied?

A.—A leaky seat at the automatic side of the tender double check valve.

426. Q.—What would cause a blow at a time that both brakes are released?

A.—A leaky triple valve slide valve, a leaky emergency valve seat, a leaky check valve case gasket or a leaky triple valve body gasket of the tender triple valve.

427. Q.—How can the source of the leak be determined?

A.—By closing the cut out cock in the brake pipe branch leading to the triple valve.

428. Q.—What is wrong if the brake then applies?

A.—The leak is from the brake pipe.

429. Q.—Which defects cause this?

A.—The leaky emergency valve seat or the leaky check valve case gasket.

430. Q.—What if the brake does not apply when the stop cock is closed?

A.—The leak is from the auxiliary reservoir pressure.

431. Q.—Which parts could be at fault?

A.—The triple valve body gasket or the slide valve.

432. Q.—How can the difference be determined?

A.—By recharging the auxiliary reservoir and making a 10-lb. brake pipe reduction.

433. Q.—What is wrong if the brake applies and the blow stops?

A.—Usually the body gasket is defective.

434. Q.—What would be wrong if the blow continued after the brake applied.

A.—The triple valve slide valve would be leaking.

435. Q.—Why are you positive of this?

A.—Because the triple valve slide valve should close the triple valve exhaust port when the brake is applied.

436. Q.—Where is a leak at the emergency exhaust port of the automatic brake valve from?

A.—From a leaky rotary valve in the automatic brake valve.

437. Q.—A leak at the brake pipe exhaust port of the automatic brake valve?

A.—Same as with the E. T. equipment.

438. Q.—What would cause either one of these brake equipments to apply with both brake valves in running position?

A.—Brake pipe leakage and an overcharged brake pipe, or brake pipe leakage in combination with a defective feed valve.

439. Q.—How is the brake cylinder leakage test made on an engine that has no straight air or independent brake?

A.—By making a 15-lb. brake pipe reduction and returning the brake valve handle to lap position.

440. Q.—Is this a fair test?

A.—Yes, if there is no brake pipe leakage in excess of 10 lbs. in five minutes.

441. Q.—Why is the test unfair if there is a greater amount of brake pipe leakage?

A.—Communication with the brake cylinders will not be closed as required by the Federal regulations.

442. Q.—What is the actual object in making but 15 lbs. brake pipe reduction, when the instructions permit of a full service reduction?

A.—If the reduction is 15 lbs. and the brake pipe leakage is 10 lbs. during the time of test, the total brake pipe reduction will have been 25 lbs. and if the brake remains applied for a period of 5 minutes under the conditions the rules of the Interstate Commerce Commission will have been complied with.

443. Q.—How is an accurate leakage test made with this brake?

A.—By attaching an air gauge to the exhaust port of the triple valve, and making a full service brake pipe reduction and returning the valve handle to running position.

444. Q.—How can the test be made in the driver brake when there is a special retaining valve or stop cock attached to the triple valve exhaust port?

A.—By closing the cock in the retainer pipe and making the full service brake application and returning the brake valve to running position.

445. Q.—How long must the brake remain applied on any locomotive or tender with communication to the brake cylinders closed?

A.—For five minutes' time.

446. Q.—What would be wrong if the brake leaked off very quickly when using a retaining valve or cut-out cock for the brake cylinder leakage test, but the brake appeared to be holding very well, when the retaining valve was not used?

A.—It would indicate that the leakage was in the piping attached to the triple valve exhaust port.

(To be continued)

Train Handling.

(*Continued from page 227, July,* 1918.)

432. Q.—What if the sticking was primarily due to low steam pressure on the locomotive boiler?

A.—The brake valve handle would be allowed to remain in release position until the steam pressure was regained.

433. Q.—Why?

A.—To add the main reservoir volume to that of the brake pipe.

434. Q.—Of what advantage would this be?

A.—Any leakage that would have a tendency to apply the brakes would then have to reduce the main reservoir pres-

Air Brake Department

Differences Between PM Brake Equipment, High Speed Brake With Type L Triple Valve and LN Equipment—Supplementary Reservoirs Should Be Used in Connection With Type L Triple Valves

Certain questions have been raised in reference to the use of supplementary reservoirs, when cars are equipped with them, it being the practice on some roads to operate type L triple valves with the supplementary reservoirs cut out. In some instances it is said that cutting them in is undesirable and that air brake troubles are encountered through attempting to operate LN equipment in mixed trains, and an explanation of the differences between these equipments has been requested.

The quick action automatic brake consists primarily of a brake cylinder for utilizing the power of compressed air and transmitting it through suitably arranged foundation brake gear to become effective in bringing the shoes in contact with the tread of the wheels, an auxiliary reservoir proportioned to the size of the brake cylinder employed and used to store a volume of compressed air for the operation of the brake piston of the brake cylinder, and a triple valve known as the type P, which consists mainly of a piston and slide valve encased in suitable bushings and which is operated through a differential in pressure obtained between the brake pipe and the auxiliary reservoir. The auxiliary reservoir is charged through the triple valve from the brake pipe pressure when the triple valve is in release position, and at this time the brake cylinder is open to the atmosphere through the triple valve slide valve. When a brake application is desired, the brake pipe pressure is reduced at a faster rate than at which it can flow back from the auxiliary reservoir through the triple valve feed groove into the brake pipe, and this results in the differential in pressure that causes the movement of the triple valve parts to admit air pressure from the auxiliary reservoir into the brake cylinder, applying the brake, the movement of the parts to application having severed communication between the brake cylinder and the atmosphere. When a release of brakes is desired, the brake pipe pressure is increased above that remaining in the auxiliary reservoir, or the auxiliary reservoir pressure may be reduced below that remaining in the brake pipe, and any such difference thus obtained moves the triple valve parts to release position, opening the brake cylinder to the atmosphere for a release of brakes and recharging the auxiliary reservoir.

The triple valve also contains a series of supplemental valves, which are operated by a sudden reduction of brake pipe pressure or when the rate of brake pipe reduction exceeds the capacity of the service port of the triple valve to expand auxiliary reservoir pressure into the brake cylinder. This causes the triple valve parts to travel their full stroke, opening the brake pipe to the brake cylinder for the continuation of the brake pipe reduction rate, or the initiation and propagation of quick action throughout the train, the intent being that the brakes on a train may become fully applied throughout in advance of any otherwise violent change of slack in the train.

To change from the quick action brake to the high speed or what is commonly known as the PM equipment, it is only necessary to increase the brake pipe pressure to 110 pounds, and add a high speed reducing valve to the brake cylinder, to limit the service braking ratio of the car to a predetermined figure. This valve also reduces the initial emergency braking ratio in a limited space of time.

The type P triple valve may be changed for one of the type L design, which is then generally known as the high-speed brake with type L triple valves; this valve contains all of the features of the P triple valve and in addition a quick service and quick recharge of the auxiliary reservoir.

If desired, a supplementary reservoir may be added to the type L triple valve, and the high-speed reducing valve removed from the brake cylinder, and the brake equipment will then be what is known as the LN, having all of the features of the high-speed brake and in addition a quick service, quick recharge of the auxiliary reservoir, high emergency brake cylinder pressure, graduated release and a safety valve feature. Commenting on these features separately:

Quick Service.—When the triple valve piston and slide valve are moved to application position, for an application of the brakes, an opening of a fixed size is made from the brake pipe to the brake cylinder, making a local brake pipe reduction at each triple valve, thus hastening the time of serial brake action throughout a train of cars, or a certain per cent of the brake pipe volume is reduced at the triple valves which would otherwise escape at the brake valve of the locomotive or at some other brake pipe opening, necessarily involving a greater element of time for the same amount of brake pipe reduction to take place. There is an unavoidable difference in time in the application of the brake on the first and last cars in a train, during service operation, and this serial hastening of the brake pipe reduction and consequently the application of the rear brakes of the train, materially assists in reducing the velocity differences between the various cars of a train, the velocity differences or differences in speed between cars of a train being the cause of shocks in trains during brake applications resulting in what is frequently termed "rough handling," and results in damage to equipment and slid flat wheels, through the fact that the change in speed through shock suddenly accelerates certain cars in a train to a greater speed than the wheel can instantly increase in speed with the brake shoe applied, with the result that the adhesion between the wheel and rail is broken and the wheel slides. The damage to equipment through shocks during brake applications cannot be disregarded when it is known an elaborate series of scientific tests have demonstrated that when the difference in speed between two cars is only one mile per hour, the impact or blow resulting is approximately equal to the weight of one car, both weights being equal, or where one car is lighter than the other the force of the impact is equal in proportion to the weight of the lighter car to that of the heavier in per cent. With trains of ten cars and both front and rear portions solidly bunched (slack out of couplings) an impact or blow of 400,000 pounds has been obtained through the influence of unequal braking effect, or through velocity differences produced by differences in time of serial brake action, hence the importance of a quick service feature which tends to minimize the shocks which are frequently of such violence that only the capacity of the draft gear to absorb shocks prevents the breaking of couplings. This is recognized to such an extent that where traffic conditions are exceedingly heavy the air brake is operated by electric current (and certain railroads are equipping steam road trains with electro-pneumatic brakes) for the express purpose of eliminating the time element incident to serial action of brakes.

Quick Recharge.—The quick recharge

manifests the greater per cent. of efficiency?

A.—The cross compound compressors.

566. Q.—Why?

A.—Because of a better location of air valves already mentioned and a reduced clearance space for the pistons and through the fact that the low pressure air piston does not work against a higher air pressure than 40 lbs. per square inch.

(To be continued.)

Train Handling.

(*Continued from page 324, October,* 1918.)

542. Q.—What advantages have the ET and LT brake equipments over former types?

A.—Locomotive and tender brake cylinder pressure will be maintained at a constant figure regardless of unequal brake piston travel and brake cylinder leakage, up to the capacity of the air compressor and the point at which the brake pistons come in contact with the non-pressure heads of the cylinders.

543. Q.—What are the principal advantages from a view point of manipulation?

A.—The locomotive and tender brake can be operated entirely independent of the train brakes, and the pressure can be graduated out of, as well as into, the brake cylinders with either one of the brake valves.

544. Q.—What are the essential parts of these equipments?

A.—An air compressor, a governor, a main reservoir, two brake valves, two duplex air gauges, two pressure controllers, suitable brake cylinders, and the usual piping, cut out cocks and hose connections.

545. Q.—What other advantages are there in the way of brake apparatus used?

A.—Less brake apparatus is required for the improved features, and but one size of operating valve and reservoir is required.

546. Q.—Is it adaptable to all classes of service?

A.—It is used in any passenger, freight or switching service without any special change in adjustment.

547. Q.—What is the name of the pipe leading from the dome or bridge pipe to the air compressor?

A.—The air pump steam pipe, or steam admission pipe.

548. Q.—What part of the equipment is in this pipe?

A.—The air pump governor.

549. Q.—Has it any other pipe connection to its steam portion?

A.—Yes, a drain or waste pipe.

550. Q.—What is the name of the pipe leading from the steam cylinder of the compressor to the locomotive cylinder saddle or the front end?

A.—The air pump exhaust pipe.

551. Q.—What is the name of the pipe leading from the air pump to the first main reservoir?

A.—The air pump discharge pipe.

552. Q.—Have the steam cylinders any other pipe connections?

A.—Sometimes steam cylinder waste or drain pipes.

553. Q.—Have the air cylinders any other pipe connections?

A.—Sometimes oil pipes leading from the locomotive lubricator.

554. Q.—What is the name of the pipe connecting the main reservoirs?

A.—The connecting pipe.

555. Q.—The name of the pipe connecting the last main reservoir and the automatic brake valve?

A.—The main reservoir pipe; usually termed the reservoir pipe.

556. Q.—The name of the pipe connecting the automatic brake valve and the feed valve?

A.—The feed valve pipe.

557. Q.—Is the feed valve pipe an essential of the equipment?

A.—No, the feed valve may be attached to the automatic brake valve bracket and require no pipe connections.

558. Q.—The pipe connecting the reducing valve and independent brake valve?

A.—The reducing valve pipe.

559. Q.—The pipe connecting the automatic brake valve, the driver brake cylinder operating valve and the angle cocks at the front and rear end?

A.—The brake pipe.

560. Q.—The pipe connecting the driver brake operating valve and the brake cylinders?

A.—The brake cylinder pipe.

561. Q.—The pipe connecting the feed valve pipe, or feed valve pipe port, in the brake valve pipe bracket, and the excess pressure governor top?

A.—The excess pressure pipe.

562. Q.—The pipe connecting the brake valve bracket and the lower connection of the excess pressure governor top?

A.—The excess pressure operating pipe.

563. Q.—The pipe connecting the maximum governor top with the main reservoir?

A.—The governor pipe.

564. Q.—The pipe connecting the main reservoir and air gauge?

A.—The reservoir gauge pipe.

565. Q.—The name of the pipe connecting the brake pipe and air gauge?

A.—The brake pipe gauge pipe.

566. Q.—The pipe connecting the equalizing reservoir and air gauge?

A.—The equalizing reservoir gauge pipe.

567. Q.—The pipe connecting the brake cylinders and the air gauge?

A.—The brake cylinder gauge pipe.

568. Q.—The pipe connecting the equalizing reservoir and the brake valve?

A.—The equalizing reservoir pipe.

569. Q.—The one connecting the main reservoir pipe and the distributing valve reservoir?

A.—The distributing valve supply pipe.

570. Q.—The pipe connecting the independent brake valve and distributing valve?

A.—The release pipe.

571. Q.—The one connecting the distributing valve and both brake valves?

A.—The application cylinder pipe.

572. Q.—The one connecting the brake valves?

A.—The release pipe branch pipe.

573. Q. The name of the pipe connecting the control valve of the LT equipment with the automatic brake valve?

A.—The control valve release pipe.

574. Q.—The one connecting the control valve and both brake valves?

A.—The control reservoir pipe.

575. Q.—The one leading from the straight air brake valve to the double check valve?

A.—The straight air pipe.

576. Q.—How do the names of the rest of the piping of the LT equipment compare with the ET?

A.—They are practically the same.

577. Q.—How are the pipes leading to the distributing valve or control valve or to the automatic brake valve from the main brake pipe referred to?

A.—As brake pipe branch pipes, specifying the part the branch leads to.

578. Q.—The pipe leading from the reservoir pipe to the pressure controllers?

A.—As the reservoir branch pipe.

579. Q. What is the difference between the independent brake of the ET and the straight air brake of the LT equipment?

A.—With the independent brake the distributing valve controls the flow of air to and from the brake cylinders, with the straight air it flows direct from the main reservoir through the straight air brake valve and double check valve to the cylinders.

580. Q.—How should an air compressor be started?

A.—With the drain cocks in the steam cylinders open and with a very light throttle or at a very slow speed.

581. Q.—When should the lubricator be started to feed the steam cylinders?

A.—As soon as the condensation has worked out of the cylinders and the drain cocks have been closed.

582. Q.—How should it be fed?

A.—With 10 or 15 drops to start with, and thereafter the feed should be regulated to from one to two drops per minute per cylinder.

583. Q.—When should the air cylinders be lubricated?

A.—As soon as the compressor is started.

584. Q.—In what quantity?

A.—8 or 10 drops per cylinder

585. Q.—Under what condition?

A.—While the compressor is working at a fair rate of speed against a low air pressure and with very hot oil.

586. Q.—What kind of oil?

A.—Valve oil.

587. Q.—Why should the oil be hot and the speed at a fair rate?

A.—So that the oil may be scattered around in the cylinder and properly lubricate it.

588. Q.—Can this be done if the air pressure is high and the speed slow?

A.—Not without using an excessive amount of oil and flooding the cylinder.

(To be continued.)

Car Brake Inspection.

(Continued from page 325, October, 1918.)

512. Q.—What is the effect of such leakage?

A.—If from the brake pipe it is a waste of air and tends to apply the brakes with a greater force than is desired, or rather tends to take the control of the brakes away from the engineman and if from the auxiliary reservoir it tends to release brakes after being applied, that is, the brake with the defective valve, and thereafter cause a drain on the brake pipe.

513. Q.—What is the effect of a leaky graduating valve?

A.—It fails to stop the flow of air from the auxiliary reservoir after the triple valve has assumed lap position and under certain conditions may result in a release of the brake.

514. Q.—In what way can it cause the brake to release?

A.—By making a reduction in the pressure in the auxiliary reservoir.

515. Q.—If the triple valve is in perfect condition will the brake release even if the graduating valve is leaking?

A.—No, as the triple valve may then be expected to move toward its release position gradually, the slide valve will eventually stop with the service port closed by the slide valve seat before the exhaust port can open.

516. Q.—Will the leaky graduating valve cause a blow at the triple valve exhaust port?

A.—Not unless it is of the slide valve kind, used in modern types of triple valves.

517. Q.—How can it sometimes be known whether a leak is from the emergency valve without an examination?

A.—By the buzzing sound accompanying the leak at the exhaust port.

518. Q.—What causes this buzzing noise?

A.—The check valve rising and falling very rapidly in supplying the leakage.

519. Q.—What causes a triple valve to fail to release?

A.—Either the brake pipe pressure has not been increased sufficiently to accomplish the release, the auxiliary reservoir has been charged to a higher pressure than that carried in the brake pipe or brake pipe pressure has passed the triple valve piston packing ring and has charged the auxiliary reservoir equal to the brake pipe, without moving the triple valve.

520. Q.—What causes a brake to fail to apply, with a light brake pipe reduction?

A.—Either excessive friction in the triple valve, leakage past the packing ring or the triple valve piston and slide valve may have moved, but the pressure has escaped from the brake cylinder through the leather or gasket.

521. Q.—What does a failure to release this sometimes result in?

A.—A slid flat wheel.

522. Q.—What defects in wheels and trucks are sometimes responsible for slid flat wheels?

A.—Wheels not bored centrally; improperly mounted on axle; wheels not perfectly round; wheels warped; unevenly chilled; improperly hung brake beams; improper length of brake beam hangers.

523. Q.—What defects of the car brakes contribute toward bad results?

A.—Unequal piston travel; brake pipe leakage; stopped up retaining valve; wrongly used triple valve; imperfect triple valve repair work; leaky emergency valve or check valve case gasket; wrongly used triple valve piston; leaky triple valve piston packing ring; too thin brake shoes.

524. Q.—What lack of observation or inspection may be responsible?

A.—Starting trains with brakes applied is the most prolific cause; retaining valve turned up; switching cars with brakes set; ice frozen in brake rigging; brake beams wedged with shoes against the wheels; clubbing down hand brakes with air brakes applied.

525. Q.—How will improper manipulation contribute?

A.—Too long a time with the brake valve in release position; allowing insufficient time for a release of brakes; failure to kick off brakes after a re-application; failure to apply brake in a manner to conform with slack conditions in train.

526. Q.—Are there any defects of the engine equipment that may result in slid flat wheels on cars?

A.—Yes, defective triple valves on engine; leaky brake cylinder; defective feed valve; no excess pressure; main reservoir full of water.

527. Q.—Is there any other prolific cause of slid flat wheels?

A.—Yes, due to slack action in trains not being properly controlled (not necessarily a fault in the manipulation) the car bodies are accelerated at a faster rate than at which the car wheels can revolve with the brake shoes retarding them and in this manner the adhesion of the wheel to the rail is broken.

528. Q.—What is the effect of a leaky check valve if found in the triple valve?

A.—It will have no effect until the brake pipe pressure is lower than the point of equalization of the brake cylinder and auxiliary reservoir, thereafter it will leak brake cylinder pressure back into the brake pipe.

529. Q.—What occurs if for any reason the brake pipe pressure reduces at a faster rate than that at which the auxiliary reservoir pressure can flow through the service port into the brake cylinder?

A.—The triple valve will travel its full stroke and assume emergency position.

530. Q.—What defects of the triple valve will cause this action?

A.—The disorders that will contribute are: A restricted service port; oil or moisture on the slide valve; a dirty or gummed up triple valve piston; excessive friction in the movement of the slide valve; a tight piston or ring; a partly closed feed groove; piston making a tight seal on the slide valve bushing; gum on piston bevel; slide valve spring catching in bushing; a weak graduating spring; a broken graduating spring; under some conditions a very short piston travel.

531. Q.—How does oil on the slide valve contribute?

A.—By making a tight seal around the edges of the slide valve and positively excluding any leakage to the under side of the slide valve which allows the entire pressure per square inch of the auxiliary reservoir pressure to be effective on the slide valve, whereas if there was a slight amount of leakage or percolation of air to the under side of the slide valve the valve would tend to be balanced and would be comparatively free in its movement.

532. Q.—How does the partly closed feed groove contribute?

A.—On a slow rate of brake pipe reduction through leakage, the triple valve piston and graduating valve may be moved to close the feed groove with the end of the piston being in contact with the slide valve, but the brake pipe reduction being insufficient to cause the further movement of the slide valve or the triple valve pistons where the feed grooves are open and free to permit of a certain amount of back flow into the brake pipe, this triple valve with the partly closed feed groove will have assumed a position in which the piston engages the slide valve with the auxiliary reservoir pressure bottled up, and if a brake pipe reduction is made at this time the valve may jump into the quick action position. On the other hand if the feed groove had been open, the piston would not have been moved until a positive brake pipe reduction had occurred and as a result it would have moved against the slide valve with sufficient force to dislodge it and move it promptly to service application position. *(To be continued.)*